

Toxicity Prediction of Peptides

Malisha Islam Tapotee
University of Central Florida
USA

ABSTRACT

The development of peptides holds immense potential for advancing drug discovery, cancer prevention, and antimicrobial therapies. Machine Learning (ML) and Deep Learning (DL) techniques are implemented to identify specific peptides with desired properties such as cytotoxicity and stability. The process begins with the utilization of ML and DL algorithms to analyze existing and customized peptide datasets and predict the properties of peptides. Subsequently, a specific peptide candidate is synthesized based on the predictions. Finally, the synthesized peptide undergoes rigorous evaluation in solving the targeted problem. In this project, several classifiers and the Super Learner Ensemble Approach are used to classify the toxin and nontoxin peptide sequences. According to the Super Learner Approach, the Accuracy score and Matthew coefficient score are 91.55 % and 83.07 % respectively.

KEYWORDS

Super Learner Ensemble, Peptide, Toxicity, XGBoost Classifier

ACM Reference Format:

Malisha Islam Tapotee. 2024. Toxicity Prediction of Peptides. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 5 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 INTRODUCTION

In this modern era, peptide is a pivotal source to prevent diseases. Peptide consists of Amino Acid. Peptide is used as a medium of drug delivery [6], shows antimicrobial activities [16], and paves the way for cancer treatment [1]. The discovery of therapeutic peptides is growing day by day [12]. Peptide consists of 50 or fewer Amino Acids (AA) with peptide bond linkage [14]. Several different Machine Learning (ML) and Deep Learning (DL) techniques [7, 12, 15] are implemented to discover the desired peptide. A peptide should be free from any kind of toxicity if we want to use it as therapeutics. Therefore, ML and DL techniques [5, 13, 14] are utilized to classify the non-toxic peptides from toxic ones for the development of medicinal peptides.

In this project, the ATSE [14] dataset is used to classify the peptide sequences into toxin and nontoxin by using ML classifiers and the Super Learner Ensemble approach. In section 2 related work is presented. Section 3 represents the overall overview of this

project, Section 4 represents the experimental part with the result. The conclusion and future work are presented in Section 5.

2 RELATED WORK

Table 1 represents some previous works on the toxicity prediction of peptides, which imply the rigorous usefulness of peptides as remedies.

The below terms are full forms of model architectures, which were used in previous studies.

- CNN : Convolutional Neural Network
- BiGRU : Bidirectional Gated Recurrent Unit
- GNN : Graph Neural Network
- BiLSTM : Bidirectional Long Short-Term Memory
- SVM : Support Vector Machines
- HMM : Hidden Markov Models
- PSI-BLAST : Position-Specific Iterative Basic Local Alignment Search Tool

3 APPROACH

Figure 1 represents the overall overview of my project, at first peptide-based 438 features extracted from the ATSE [14] balanced dataset. Then the dataset is trained by using Stratified Cross Validation for multiple models. Among 27 classifiers, XGBoost Classifier [3] performs best, then by using this model important features are selected based on Akaike Information Criteria (AIC) [6, 11]. By using these 53 selected features the hyperparameters of XGBoost Classifier are tuned. Finally, the prediction on the test data is computed by using the Super Learner Ensemble Approach and Optimized XGBoost. Then the output is compared between these two approaches based on the evaluation metrics. The output of the Super Learner [10] Ensemble Approach is also compared with the previous study on this dataset [14]. The reason behind using the Super Learner Ensemble approach it ensembles the prediction of multiple models. Hence, this approach can perform better on unseen data than any single model [10].

3.1 Dataset Description

The dataset is given in fasta file format of 3864 Peptide sequences. Then the file is converted to CSV file format. No null value is found. No peptide sequence is duplicated. The dataset is balanced. [14] 1932 peptides are toxins and the rest 1932 are nontoxins. Since the target variable is categorical, this variable is converted to numerical. 0 is assigned for toxin, and 1 is assigned for nontoxin.

3.2 Feature Explanation

A total of 483 features are implemented and no null value is found. Table 2 represents the description of features [6], which are obtained from peptide sequences. The number of total extracted features from each peptide sequence is also mentioned.

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Conference acronym 'XX, June 03–05, 2024, Woodstock, NY

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ACM ISBN 978-1-4503-XXXX-X/18/06

<https://doi.org/XXXXXXX.XXXXXXX>

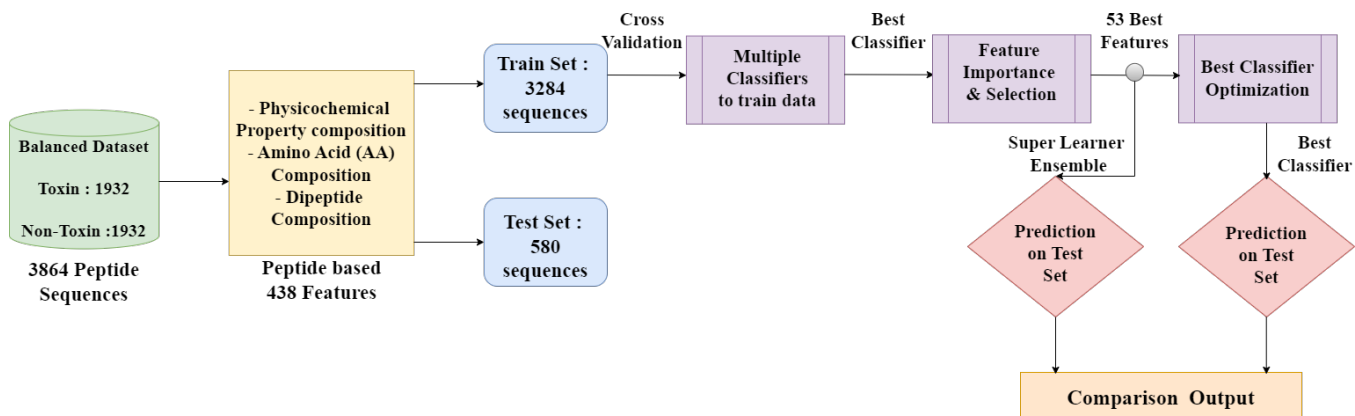


Figure 1: Overall Overview of Project.

Table 2: Feature Description

Feature Name	Description	Total Extracted Features
Weight	Summation of weights of each amino acid of a peptide	1
Tiny	Percentage of 6 Amino Acids ("ACGSTU")	1
Small	Percentage of 11 Amino Acids ("ABCDGNPSTUV")	1
Aliphatic	Percentage of 4 Amino Acids ("AILV")	1
Aromatic	Percentage of 4 Amino Acids ("FHWY")	1
NonPolar	Percentage of 12 Amino Acids ("ACFGILMPUVWY")	1
Polar	Percentage of 10 Amino Acids ("DEHKNQRSTZ")	1
Charged	Percentage of 7 Amino Acids ("BDEKRRHZ")	1
Basic	Percentage of 3 Amino Acids ("HKR")	1
Acidic	Percentage of 3 Amino Acids ("BDEZ")	1
Isoelectric Point	pH level of 0 net charge	1
Hydrophobicity	Grand average of hydropathicity of peptide sequence	1
Net Charge	Amount of charge at pH = 7	1
Boman Index	Interaction between peptides	1
Instability Index	Estimation of the peptide stability in test tube	1
Fraction of Secondary Structure	Percentage of Sheet, Turn, and Helix	3
Amino Acid (AA) Composition	Percentage of each Amino Acid	20
Dipeptide Amino Acid (AA) Composition	Percentage of every Dipeptide Amino Acid	400

4 EXPERIMENT

In this section, all analyses and output are presented.

4.1 Dataset Split

At first, the dataset is split into train, validation, and test sets. To get the generalization of data, stratified cross-validation [2] is applied. Among 3684 samples, 15% (580) samples are used as the Test set. Then from 3284 remaining samples 10% (329) samples are used as the Validation set and the remaining samples are used as the Train set. Figure 2 represents the distribution of the dataset.

DATASET SAMPLE DISTRIBUTION

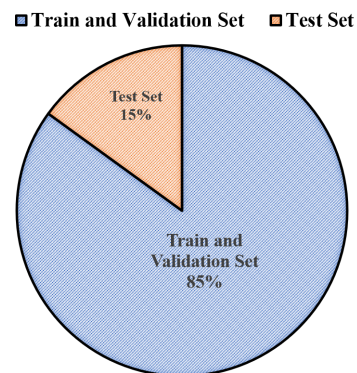


Figure 2: Distribution of Peptide Sequences into Train, Validation, and Test set.

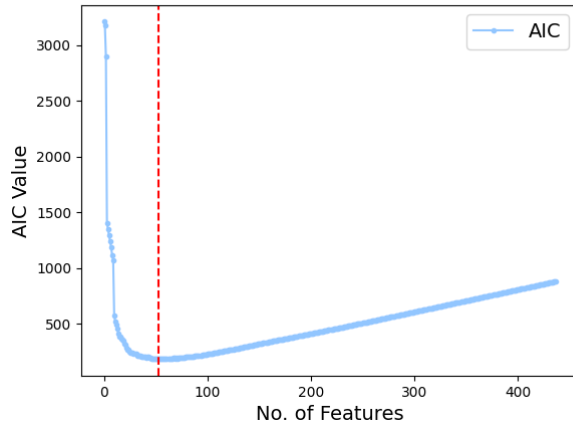
Table 3: Output of Evaluation Metrics on Validation data for top 5 models

Model Name	ACC	MCC
XGBClassifier	94 %	88 %
ExtraTreesClassifier	93 %	86 %
LGBMClassifier	93 %	86 %
RandomForestClassifier	92 %	85 %
BaggingClassifier	91 %	82 %

4.2 Train Data with Multiple Classifiers

Python library Lazypredict Classifier is used to train the data with multiple-type classifiers. At first, stratified 5-fold validation is used. Lazypredict gives an output of nearly 27 classifiers. Among them, the top 5 are selected from 5 folds for further steps. This step is only implemented on the train and validation data. Matthew coefficient score (MCC) [4] and Accuracy score (ACC) of 5 top models are provided in Table 3. Among all models, the output of these 4 models XGBClassifier, ExtraTreesClassifier, LGBMClassifier, and RandomForestClassifier is approximately near. XGBClassifier is chosen due to a slightly better MCC.

4.3 Feature Importance and Selection

**Figure 3: Selection of Important Features according to the lowest AIC value.**

To extract the important features from 438 features, 10000 trees are used of XGBoost Classifier. Then The cumulatively features are used from high to low rank to evaluate the XGBoost Classifier based of AIC value [11], and at this time 1000 trees are used. Finally, 53 features are found to be most important on the basis of the lowest AIC. The classifier gives better accuracy when the descending order important features are 53 and AIC is the lowest.

$$AIC = 2 * k - 2 * \ln(L) \quad (1)$$

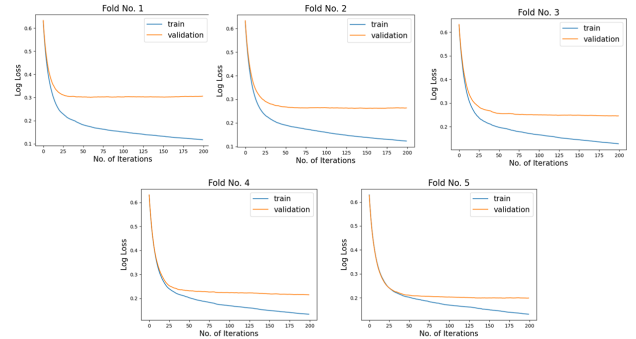
Table 4: Hyperparameters Combination of XGBoost Classifier

Hyperparameter Name	Values
Maximum Depth	[4, 5, 6]
Minimum Child Weight	[4, 5, 6]
No. of Trees	[10, 50, 100, 200, 1000]
Learning Rate	[0.01, 0.1, 0.3, 0.5]

Equation (1) represents the AIC formula. L represents the binary log loss function, and m is the number of cumulative features each time.

4.4 Hyperparameters Tuning

By using the most important 53 features from the previous step, Hyperparameters are tuned for the XGBoost Classifier. Table 4 represents the hyperparameters list and combination of values to find out the best parameter. The best accuracy happened when the parameters were set as 6, 5, 200, and, 0.1 for Maximum Depth, Minimum Child Weight, No. of Trees, and Learning Rate respectively. By using this combination, the learning curve is plotted for 5-fold cross-validation. The curves depict that the classifier is also not overfitted for this combination of hyperparameters.

**Figure 4: 5 fold Learning Curve of XGBoost Classifier.**

4.5 Super Learner Ensemble Approach

Super Learning [10] is a stacking application where all of the models are combined. At first, prediction is calculated by using base models through stratified cross-validation. Then the prediction from each classified is used as input for the meta-model. Then final prediction is calculated by using the meta-model. Figure 5 represents the Super Learner Approach. For this project, to get the prediction on test data, the linear classifier called Logistic Regression is used as the meta-model. And as base models 6 models (Logistic Regression, XGBoost Classifier, ExtraTreesClassifier, RandomForest Classifier, Light Gradient Boosting Machine, and Support Vector Machine) are implemented. Except for Logistic Regression, another 5 base models are chosen according to subsection 4.2.

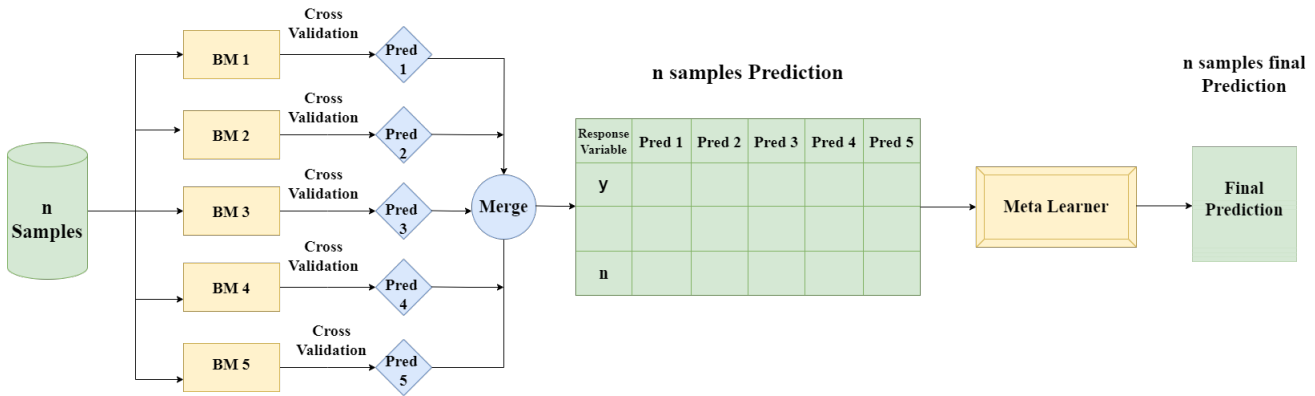


Figure 5: Diagram of Super Learner Ensemble Approach.

4.6 Evaluation Metrics

After getting the prediction on the test dataset by using the Super Learner Ensemble approach and XGBoost classifier, it is found that according to the evaluation metrics Super Learner Ensemble performs better than XGBoost Classifier. Figure 6 represents the comparison of metrics between these two approaches.

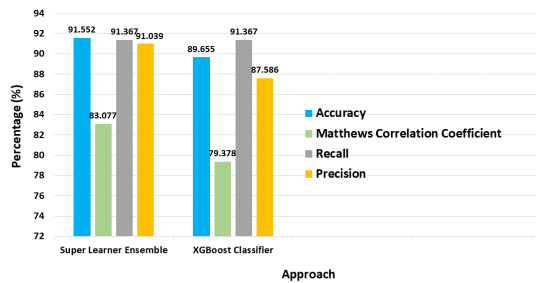


Figure 6: Evaluation Metrics Comparison between Super Learner Ensemble and XGBoost Classifier.

According to Figure 7, Both approach correctly classifies 254 samples as nontoxin (label 1). Super Learner Ensemble performs better XGBoost for toxic sequence identification.

$$rootterm = \sqrt{((TP + FP) * (TP + FN) * (TN + FP) * (TN + FN))} \quad (2)$$

$$MCC = (TP * TN - FP * FN) / rootterm \quad (3)$$

Equation (3) represents the formula of MCC. Here, TP = True Positive (Truly identified toxin) TN = True Negative (Truly identified nontoxin) FP = False Positive (Toxin one is identified as nontoxin) FN = False Negative (Nontoxin one is identified as a toxin)

The high value of MCC [4] is obtained when good prediction is introduced on the 4 groups of the confusion matrix. Since MCC

Table 5: Comparison between Super Learner Ensemble and Previous Study on this dataset

Method	Accuracy	MCC
Super Learner Ensemble (this study)	0.9155	0.8307
ATSE [14]	0.952	0.903

is a combination of 4 groups, it is used here. For Super Learner Ensemble, MCC is greater than XGBoost Classifier.

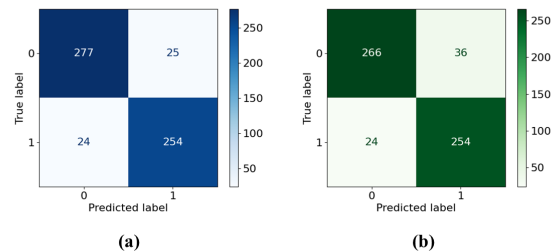


Figure 7: Confusion Matrix (a) Super Learner Ensemble , and (b) XGBoost Classifier.

4.7 Comparison with previous study

It is acknowledged from Table 5 that the overall result of the Super Learner Ensemble Approach is a little bit lower than the ATSE [14] study. Since the ATSE study used GNN and a Hybrid network of CNN and biLSTM, due to the deep architecture of the network, it may have a chance to get a better output than the Super Learner approach. Overall, the output of Super Learner is satisfactory.

5 CONCLUSION AND FUTURE WORK

As future works, planning to calculate more physiochemical peptide properties and evaluate the model. Overall performance of

superlearner is not bad, which implies it can be used in further process. Different deep learning based model architecture can be used as future work. If we classify the peptide property properly, we can synthesize it in laboratory easily which reduces our time and cost. Finally, as a part of future work by using ML find out the best peptide which can be used for as antimicrobial one or cancer prevention.

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A RESEARCH METHODS

Python Programming Language is used in most cases. The isoelectric and Bio.SeqUtils.ProtParam packages of Python are used to analyze the peptide sequences. To calculate the net charge and dipeptide amino acid composition, the packages peptides and protr and of R language is used respectively. VScode platform is used to implement the project.